

J. Benjamin Hutchinson

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Academic Positions

2018-present	Assistant Professor of Psychology, University of Oregon
2017-2018	Assistant Professor of Psychology, Northeastern University Visiting Scientist, Martinos Center, Massachusetts General Hospital

Education and Training

2011-2016	Postdoctoral Associate, Princeton University, Neuroscience Institute Advisor: Nicholas Turk-Browne
2006-2011	PhD, Stanford University, Psychology (Neuroscience Area) Advisor: Anthony Wagner
2000-2004	BA, University of Pennsylvania, Psychology Magna Cum Laude, Phi Beta Kappa, Dean's List (2001-2004) Advisor: John Trueswell

Research Support and Other Funding

2018-2021	NCS-FO: Leveraging Deep Probabilistic Models to Understand the Neural Bases of Subjective Experience Role: Co-PI (PI: Jan-Willem van de Meent)
2020	University of Oregon Data Science Initiative Seed Funding: Relating learning and intelligence across artificial and human agents Role: PI (joint with Daniel Lowd)
2011-2014	NRSA F32 (NEI): Guidance of Attention from Plasticity in Ventral Visual Cortex Role: PI (Co-Sponsors: Nicholas Turk-Browne, Kenneth Norman)

Publications

Allen EJ, St-Yves G, Wu Y, Breedlove JL, Dowdle LT, Caron B, Pestilli F, Charest I, Hutchinson JB, Naselaris T*, Kay K* (2021). A massive 7T fMRI dataset to bridge cognitive and computational neuroscience. *bioRxiv preprint*: 2021.02.22.432340.

- Azari B, Westlin C, Satpute AB, Hutchinson JB, Kragel PA, Hoemann K, Khan Z, Wormwood JB, Quigley KS, Erdogmus D, Dy J, Brooks DH, Barrett LF (2020). Comparing supervised and unsupervised approaches to emotion categorization in the human brain, body, and subjective experience. *Scientific Reports*, 10:20284.
- Farnoosh A, Rezaei B, Sennesh EZ, Khan Z, Dy J, Satpute A, Hutchinson JB, van de Meent J-W, Ostadabbas S (2020). Deep Markov Spatio-Temporal Factorization. *arXiv preprint: arXiv:200309779*
- Kumar M, Anderson MJ, Antony JW, Baldassano C, Brooks PP, Cai MB, Chen P-HC, Ellis CT, Henselman-Petrusek G, Huberdeau D, Hutchinson JB, Li PY, Lu Q, Manning JR, Mennen AC, Nastase SA, Richard H, Schapiro AC, Schuck NW, Suo D, Turek JS, Vo VA, Wallace G, Wang Y, Zhang H, Zhu X, Capotă M, Cohen JD, Hasson U, Li K, Ramadge PJ, Turk-Browne NB, Willke TL, Norman KA (2020) BrainIAK: The Brain Imaging Analysis Kit. *OSF preprint: 10.31219/osf.io/db2ev*
- Khan Z*, Wang Y*, Sennesh E, Dy J, Ostadabbas S, Van De Meent JW, Hutchinson JB[†], Satpute AB[†] (2020). A computational neural model for mapping degenerate neural architectures. *bioRxiv preprint: 2020.11.13.382192*
- Sennesh E, Khan Z, Wang Y, Hutchinson JB, Satpute AB, Dy J, van de Meent J-W (2020). Neural Topographic Factor Analysis for fMRI data. *Advances in Neural Information Processing Systems* 33.
- Hutchinson JB, Barrett LF (2019). The power of predictions: an emerging paradigm for psychological research. *Current Directions in Psychological Science*. 28, 280-291.
- Chen D, Hutchinson JB (2018). What is memory-guided attention? How past experiences shape selective visuospatial attention in the present. In: *Current Topics in Behavioral Neurosciences*. Springer, Berlin, Heidelberg.
- Fan J*, Hutchinson JB*, Turk-Browne NB (2016). When past is present: substitutions of long-term memory for sensory evidence in perceptual judgments. *Journal of Vision*, 16, 1-12.
- Hutchinson JB, Pak SS, Turk-Browne NB (2016). Biased competition during long-term memory formation. *Journal of Cognitive Neuroscience*, 28, 187-197.
- Gonzalez A, Hutchinson JB, Uncapher MR, Chen J, LaRocque KF, Foster BL, Rangarajan V, Parvizi J, Wagner AD (2015). Electrocorticography reveals the temporal dynamics of posterior parietal cortical activity during recognition memory decisions. *Proceedings of the National Academy of Sciences*, 112, 11066-11071.
- Hutchinson JB, Uncapher MR, Wagner AD (2015). Increased functional connectivity between dorsal posterior parietal and ventral occipitotemporal cortex during uncertain memory decisions. *Neurobiology of Learning and Memory*, 117, 71-83.
- Farah MJ, Hutchinson JB, Phelps EA, Wagner AD (2014). Functional MRI-based lie detection: scientific and societal challenges. *Nature Reviews Neuroscience*, 15, 123-131.

- Hutchinson JB, Uncapher MR, Weiner KS, Bressler DW, Silver MA, Preston AR, Wagner AD (2014). Functional heterogeneity in posterior parietal cortex across attention and episodic memory retrieval. *Cerebral Cortex*, 24, 49-66.
- Hutchinson JB, Turk-Browne NB (2012). Memory-guided attention: control from multiple memory systems. *Trends in Cognitive Sciences*, 16, 576-579.
- Uncapher MR, Hutchinson JB, Wagner AD (2011). Dissociable effects of top-down and bottom-up attention during episodic encoding. *Journal of Neuroscience*, 31, 12593-12603.
- Adamson MM, Hutchinson JB, Shelton A, Wagner AD, Taylor JL (2011) Reduced hippocampal activity during encoding in cognitively normal adults carrying the APOE ϵ 4 allele. *Neuropsychologia*, 49, 2448-2455.
- Uncapher MR, Hutchinson JB, Wagner AD (2010). A roadmap to brain mapping: toward a functional map of human parietal cortex. *Neuron*, 67, 5-8.
- Preston AR, Bornstein AM, Hutchinson JB, Gaare ME, Glover GH, Wagner AD (2010). High-resolution fMRI of content-sensitive subsequent memory responses in human medial temporal lobe. *Journal of Cognitive Neuroscience*, 22, 156-173.
- Hutchinson JB*, Uncapher MR*, Wagner AD (2009). Posterior parietal cortex and episodic retrieval: convergent and divergent effects of attention and memory. *Learning and Memory*, 16, 343-356.

*, † = equal contributions

Conference Presentations

- Li PY, Wang Y, Turk-Browne NB, Hutchinson JB (March 2021). Characterizing external and internal attention: Functional connectivity reveals multiple interacting processing streams. *Cognitive Neuroscience Society*, Virtual Conference.
- Price MH, Allen EJ, Wu Y, Naselaris T, Kay KN, Hutchinson JB (March 2021). MTL subregions differentially track repetition and recency across large timescales. *Cognitive Neuroscience Society*, Virtual Conference.
- Zou F, Allen EJ, Wu Y, Charest I, Naselaris T, Kay KN, Hutchinson JB, DuBrow S (March 2021). For the when: The role of the medial temporal lobe in encoding long-timescale temporal information. *Cognitive Neuroscience Society*, Virtual Conference.
- Price MH, Allen EJ, Wu Y, Naselaris T, Kay KN, Hutchinson JB (November 2020). Long-term continuous recognition: Characterizing repetition effects at intervals spanning from seconds to months. *Psychonomics Society*, Virtual Conference.
- Allen EJ, Wu Y, Hutchinson JB, Naselaris T, Kay KN (June 2020). The natural scenes dataset: A yearlong ultra-high field whole-brain human fMRI visual perception and memory study. *Vision Sciences Society*, Virtual Conference.

- Wang Y, Khan A, Sennesh E, Dy J, Satpute A, Hutchinson JB, van de Meent JW (May 2020). Modeling degenerate neural architecture using Neural Topographic Factor Analysis. *Cognitive Neuroscience Society*, Virtual Conference.
- Allen EJ, Wu Y, Hutchinson JB, Naselaris T, Kay KN (Oct 2019). The natural scenes dataset: massive, high-quality whole-brain 7T fMRI measurements during visual perception and memory. *Society for Neuroscience*, Chicago, IL.
- Biba T, Singh I, Hutchinson JB (March 2019). The neural basis of internal attention: characterizing attentional orienting along a memory array. *Cognitive Neuroscience Society*, San Francisco, CA.
- Collins JA, Dickerson BC, Hutchinson JB (March 2019). Data-driven analysis of whole-brain connectivity reveals post-encoding network dynamics. *Cognitive Neuroscience Society*, San Francisco, CA.
- Singh I, Hutchinson JB (May 2018). Orienting of attention along a sequentially presented memory array. *Association for Psychological Science*, San Francisco, CA.
- Chen D, Hutchinson JB (May 2018). The influence of temporal structure on visual search: how prediction shapes top-down and bottom-up attention. *Vision Sciences Society*, St. Pete Beach, FL.
- Gonzalez A, Hutchinson JB, Uncapher MR, Parvizi J, Wagner AD (Nov 2017). Oscillatory states in human parietal cortex predict subsequent memory performance: evidence from ECoG recordings. *Society for Neuroscience*, Washington, DC.
- Suo DC, Hutchinson JB, deBettencourt M, Mennen A, Wang Y, Willke T, Turk-Browne NB, Norman KA, Cohen JD, Li K (Nov 2017). Real-time fMRI analysis in the cloud. *Society for Neuroscience*, Washington, DC.
- Hutchinson JB, Wang Y, Turk-Browne NB (Nov 2016). Disentangling remembered and perceived information in the full correlation matrix of human brain activity. *Society for Neuroscience*, San Diego, CA.
- Hutchinson JB, Wang Y, Turk-Browne NB (May 2016). Overlap and separation of remembered and perceived visual information in the human medial temporal lobe. *Vision Sciences Society*, St. Pete Beach, FL.
- Hutchinson JB, Turk-Browne NB (Oct 2015). Object-based competition during long-term memory encoding. *Society for Neuroscience*, Chicago, IL.
- Hutchinson JB, Wang Y, Turk-Browne NB (Nov 2014). Decoding the locus of attention from the full correlation matrix of the human brain. *Society for Neuroscience*, Washington, DC.
- Gonzalez A, Hutchinson JB, LaRocque KF, Chen J, Foster BL, Rangarajan V, Parvizi J, Wagner AD (Nov 2014). Intracranial evidence of parietal cortex involvement in the encoding of episodic memories. *Society for Neuroscience*, Washington, DC.
- Hutchinson JB, Turk-Browne NB (May 2014). Guidance of object-based attention from neural signatures of memory. *Vision Sciences Society*, St. Pete Beach, FL.

- Pak SS, Hutchinson JB, Turk-Browne NB (May 2014). Intuitive statistics from graphical representations of data. *Vision Sciences Society*, St. Pete Beach, FL.
- Fan JE, Hutchinson JB, Turk-Browne, NB (Nov 2013). Incidental expression of visual long-term memory in online perception. *Psychonomic Society*, Toronto, ON.
- Hutchinson JB, Turk-Browne, NB (Nov 2013). Interacting signals of memory during competitive encoding. *Society for Neuroscience*, San Diego, CA.
- Hutchinson JB, Pak SS, Turk-Browne NB (Jun 2013). Competitive dynamics of the encoding of novel information. *Manhattan Area Memory Meeting*, New York, NY.
- Gonzalez A, Hutchinson JB, Uncapher MR, Chen J, LaRocque KF, Foster BL, Rangarajan V, Parvizi J, Wagner AD (Apr 2013). Oscillatory dynamics in the human parietal cortex during recognition memory decisions. *Cognitive Neuroscience Society*, San Francisco, CA.
- Hutchinson JB, Pak SS, Turk-Browne NB (Oct 2012). Memory-guided attention: how prior experience influences encoding of novel information. *Society for Neuroscience*, New Orleans, LA.
- Gonzalez-Barbosa A, Hutchinson JB, Uncapher MR, Parvizi J, Wagner AD (Oct 2012). Decoding of recognition memory states from ECoG local field potentials in human parietal cortex. *Society for Neuroscience*, New Orleans, LA.
- Hutchinson JB, Pak SS, Turk-Browne NB (May 2012). When old meets new: repetition enhances encoding of competing novel items. *Vision Sciences Society*, Naples, FL.
- Uncapher MR, DuBrow S, Hutchinson JB, Wagner AD (Nov 2011). Temporal interplay of attention and memory during associative retrieval. *Society for Neuroscience*, Washington DC.
- Uncapher MR, DuBrow S, Hutchinson JB, Wagner AD (Nov 2011). Associative retrieval exhibits a dynamic temporal interplay between attention and memory. *Bay Area Memory Meeting*, Davis, CA.
- Hutchinson JB, Uncapher MR, Wagner AD (Aug 2010). Effects of decision framing on episodic retrieval. *Bay Area Memory Meeting*, Stanford, CA.
- Hutchinson JB, Uncapher MR, Wagner AD (Apr 2010). Parietal contributions to episodic retrieval: effects of memory and decision criteria. *Cognitive Neuroscience Society*, Montreal, Canada.
- Uncapher MR, DuBrow S, Hutchinson JB, Wagner AD (Apr 2010). Remembering to attend: separating memory and attention signals in the posterior parietal cortex. *Cognitive Neuroscience Society*, Montreal, Canada.
- Hutchinson JB, Uncapher MR, Wagner AD (Mar 2009). Contributions of parietal cortex to attention and memory: divergent processes of visual attention and episodic retrieval. *Cognitive Neuroscience Society*, San Francisco, CA.

Hutchinson JB, Uncapher MR, Bressler DW, Silver MA, Wagner AD (Apr 2008). Episodic memory retrieval and parietal cortex: relating memory to topographic maps of visuo-spatial attention and reflexive orienting. *Society for Neuroscience*, Washington DC.

Uncapher MR, Hutchinson JB, Wagner AD (Apr 2008). Attentional capture increases mnemonic capture: episodic encoding is modulated by the ventral parietal attention system. *Society for Neuroscience*, Washington, DC.

Hutchinson JB, Uncapher MR, Bressler DW, Silver MA, Wagner AD (Aug 2008). Parietal contributions to episodic memory retrieval and visuo-spatial attention. *Bay Area Memory Meeting*, Davis, CA.

Uncapher MR, Hutchinson JB, Wagner AD (Aug 2008). From negative to positive: harnessing the ventral attentional system to promote rather than detract from episodic encoding. *Bay Area Memory Meeting*, Davis, CA.

Hutchinson JB, Cristo Y, Preston AR, Wagner AD (Apr 2008). Frontal and parietal correlates of graded recollection during episodic retrieval. *Cognitive Neuroscience Society*, San Francisco, CA.

Invited Talks

CNS Seminar, University of Arizona, November, 2019

Cognitive Neuroscience Seminar, University of Texas, Austin, April, 2019

Psychology Department, University of Oregon, March, 2018

Neuroimaging & Neuropsychology Lecture Series, Boston VA Hospital, December, 2017

Brain, Behavior, and Cognition Seminar, Boston University, September, 2017

Network Science Institute, Northeastern University, May, 2017

Psychology Department, Florida Atlantic University, March, 2016

Psychology Department, University of California, Riverside, March, 2016

Psychology Department, Northeastern University, February, 2016

Psychology Department, Tufts University, November 2014

Ebbinghaus Empire Meeting, University of Toronto, November, 2014

Cognitive Research Seminar, Princeton University, October, 2011

Friday Cognitive Seminar, Stanford University, February, 2011

Professional Memberships

Cognitive Neuroscience Society, Memory Disorders Research Society, Society for Neuroscience, Vision Sciences Society

Other Research Experience

2005-2006	Lab Manager, Stanford University Principal Investigator: Anthony Wagner
2004-2005	Research Assistant, Stanford University Principal Investigator: John Gabrieli
2003-2004	Undergraduate Research Assistant, University of Pennsylvania Principal Investigator: John Trueswell

Teaching Experience

2020-21	Cognitive Neuroscience, Instructor
2019-21	Cognitive Psychology, Instructor
2019	Graduate Seminar, Measuring Memories
2019, 21	Graduate Course on Cognitive Neuroscience, Co-Instructor
2018	Graduate Seminar on Cognition, Instructor
2016	Introduction to Cognitive Neuroscience, Lab Instructor
2009	Introduction to Cognitive Neuroscience, Teaching Assistant
2008	Introduction to Learning and Memory, Teaching Assistant
2008	Introductory Statistics, Teaching Assistant
2008	MATLAB and Psychtoolbox for the Behavioral Sciences, Co-Instructor
2008	Introduction to Cognitive and Comparative Neuroscience, Co-Instructor
2007	Cellular Neuroscience, Teaching Assistant

Mentoring Experience

2020-	David Han (UO lab manager)
2020-	Nathan Young (UO graduate student)
2019-	Mason Price (UO post-doc)
2019-	Futing Zou (UO graduate student; jointly advised with Sarah DuBrow)
2019-	Peeta Li (UO graduate student)
2017-	Danlei Chen (Northeastern graduate student)
2017-2020	Thomas Biba (Northeastern/UO lab manager)

2017-2018	Inderdeep Singh (Northeastern post-doc)
2015	Rachel Klebanov (Princeton undergrad)
2013	Annie Chen (CMU undergrad)
2013	Max Luo (area high school student)
2011-2015	Sarah Pak (Princeton undergrad)
2010	Tim Jausovec (Stanford undergrad)
2009	Nandini Datta (UCSD undergrad)
2007-2008	Valerie Ross (Stanford undergrad)
2007	Michael King (Stanford undergrad)
2007	Maria Corona (CSU undergrad)
2007	Yudy Cristo (Leadership Alliance Research Mentor program)

Ad Hoc Reviewing

Acta Psychologica • Attention, Perception and Psychophysics • Biological Psychology • Brain Research • Brain Structure and Function • Cerebral Cortex • Cognitive, Affective, and Behavioral Neuroscience • Cognitive Neuroscience • Cognitive Science • Cortex • eLife • Experimental Brain Research • Frontiers in Human Neuroscience • Human Brain Mapping • Journal of Cognitive Neuroscience • Journal of Experimental Psychology: General • Journal of Experimental Psychology: Human Perception and Performance • Journal of Experimental Psychology: Learning, Memory, and Cognition • Journal of Neuroscience • Nature Reviews Neuroscience • Neurobiology of Aging • NeuroImage • NeuroImage: Clinical • Neuropsychologia • PLOS Biology • PLOS One • Progress in Neurobiology • Psychological Research • Psychological Science • Psychonomic Bulletin and Review • Scientific Reports • Social, Cognitive, and Affective Neuroscience • Trends in Neurosciences • Visual Cognition